Introduction to Quantum Mechanics
Second Edition

David J. Griffiths

Solutions by David A. Lee

All errors, typographical and substantive, and other offenses, are entirely my own.

Table of Contents

1 - The Wave Function **3**
1.3 - Probability 3
1.4 - Normalization 7
1.5 - Momentum 11
1.6 - The Uncertainty Principle 14
Further Problems for Chapter 1 16

1 - The Wave Function

1.3 - Probability

Problem: 1.1

For the distribution of ages in Section 1.3.1:

(a) Compute $\langle j^2 \rangle$ and $\langle j \rangle^2$.

(b) Determine Δj for each j, and use Equation 1.11 to compute the standard deviation.

(c) Use your results in (a) and (b) to check Equation 1.12.

The data is

$$\begin{aligned} N(14) &= 1 \\ N(15) &= 1 \\ N(16) &= 3 \\ N(22) &= 2 \\ N(24) &= 2 \\ N(25) &= 5 \end{aligned}$$

(a) Our sample size is $N = 14$. Compute

$$\begin{aligned} \langle j^2 \rangle &= \sum_j j^2 P(j) \\ &= \frac{14^2 \cdot 1 + 15^2 \cdot 1 + 16^2 \cdot 3 + 22^2 \cdot 2 + 24^2 \cdot 2 + 25^2 \cdot 5}{14} \\ &= \frac{3217}{7} \approx 459.6 \\ \langle j \rangle^2 &= \left[\sum_j j P(j) \right]^2 \\ &= \left(\frac{14 \cdot 1 + 15 \cdot 1 + 16 \cdot 3 + 22 \cdot 2 + 24 \cdot 2 + 25 \cdot 5}{14} \right)^2 \\ &= \left(\frac{147}{7} \right)^2 = 21^2 = 441 \end{aligned}$$

Taking the difference produces the variance:

$$\sigma^2 = \langle j^2 \rangle - \langle j \rangle^2 = \frac{3217}{7} - 441 = \frac{130}{7} \approx 18.6$$

(b) Compute

$$\begin{aligned}\sigma^2 &= \left\langle (\Delta j)^2 \right\rangle = \sum_j (\Delta j)^2 \, P(j) \\ &= \frac{(14-21)^2 \cdot 1 + (15-21)^2 \cdot 1 + (16-21)^2 \cdot 3 + (22-21)^2 \cdot 2}{14} \\ &\quad + \frac{(24-21)^2 \cdot 2 + (25-21)^2 \cdot 5}{14} \\ &= \frac{130}{7} \approx 18.6 \\ \sigma &= \sqrt{130/7} \approx 4.3\end{aligned}$$

(c) The computations confirm $\left\langle (\Delta j)^2 \right\rangle = \langle j^2 \rangle - \langle j \rangle^2$. Rooting gives us standard deviation $\sqrt{130/7} \approx 4.3$.

Problem: 1.2

(a) Find the standard deviation of the distribution in Example 1.1.

(b) What is the probability that a photograph, selected at random, would show a distance x more than one standard deviation away from the average?

(a) Using formula (1.12),

$$\begin{aligned}\langle x^2 \rangle &= \int_0^h x^2 \frac{1}{2\sqrt{hx}} \, dx \\ &= \frac{1}{2\sqrt{h}} \left(\frac{2}{5} x^{5/2} \right) \Big|_0^h \\ &= \frac{h^2}{5}\end{aligned}$$

And since $\langle x \rangle = h/3$, we have $\langle x \rangle^2 = h^2/9$. Then compute

$$\sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{h^2}{5} - \frac{h^2}{9}} = \sqrt{\frac{4h^2}{45}} = \frac{2h}{3\sqrt{5}}$$

(b) We integrate the probability density over interval: $\left[h\left(1/3 - 2/3\sqrt{5}\right), h\left(1/3 + 2/3\sqrt{5}\right)\right]$[1] for the intermediate step:

$$\begin{aligned}\int_{h\left(1/3-2/3\sqrt{5}\right)}^{h\left(1/3+2/3\sqrt{5}\right)} \frac{1}{2\sqrt{hx}} \, dx &= \frac{1}{2\sqrt{h}} \left(2x^{1/2} \right) \Big|_{h\left(1/3-2/3\sqrt{5}\right)}^{h\left(1/3+2/3\sqrt{5}\right)} \\ &= \sqrt{\underbrace{\frac{1}{3} + \frac{2}{3\sqrt{5}}}_{A}} - \sqrt{\underbrace{\frac{1}{3} - \frac{2}{3\sqrt{5}}}_{B}}\end{aligned}$$

[1] The bounds fall within $[0, h]$, as $h\left(1/3 - 2/3\sqrt{5}\right) \approx 0.0352h$ and $h\left(1/3 + 2/3/\sqrt{5}\right) \approx 0.6315h$.

Observing that $\left(\sqrt{A}-\sqrt{B}\right)^2 = A + B - 2\sqrt{AB}$, compute and take the square root to simplify to

$$\frac{2}{3} - 2\sqrt{\frac{1}{9} - \frac{4}{45}} = \frac{2}{3} - \frac{2}{3\sqrt{5}} \quad \Longrightarrow \quad \sqrt{A} - \sqrt{B} = \sqrt{\frac{2}{3} - \frac{2}{3\sqrt{5}}} \approx 0.607$$

Lastly, subtracting from 1 yields a probability of ≈ 0.4.

Problem: 1.3

Consider the **gaussian** distribution

$$\rho(x) = Ae^{-\lambda(x-a)^2}$$

where A, a, and λ are positive real constants. (Look up any integrals you need.)

(a) Use Equation 1.16 to determine A.

(b) Find $\langle x \rangle$, $\langle x^2 \rangle$, and σ.

(c) Sketch the graph of $\rho(x)$.

(a) From Equation 1.16, we require

$$\int_{-\infty}^{\infty} Ae^{-\lambda(x-a)^2}\, dx = 1$$

For simplicity, let $u = x - a$. Then use the result

$$\int_{-\infty}^{\infty} e^{-\lambda x^2}\, dx = \sqrt{\frac{\pi}{\lambda}}$$

to deduce

$$\int_{-\infty}^{\infty} Ae^{-\lambda u^2}\, du = A\sqrt{\frac{\pi}{\lambda}} = 1 \quad \Longrightarrow \quad A = \sqrt{\frac{\lambda}{\pi}}$$

The intuition behind this substitution, which horizontally shifts $\rho(x)$ by a, is that the area underneath the Gaussian is independent of horizontal translation.

(b) To compute $\langle x \rangle$, make the same substitution $u = x - a$. Then $x = u + a$, and we compute

$$\langle x \rangle = \int_{-\infty}^{\infty} Axe^{-\lambda(x-a)^2}\, dx = A\left[\int_{-\infty}^{\infty} ue^{-\lambda u^2}\, du + \int_{-\infty}^{\infty} ae^{-\lambda u^2}\, du\right] = a$$

By antisymmetry of the first integrand about $x = a$, the first term evaluates to zero (put differently, odd in u). The second is simply the normalized Gaussian with a scale factor, leaving only a.

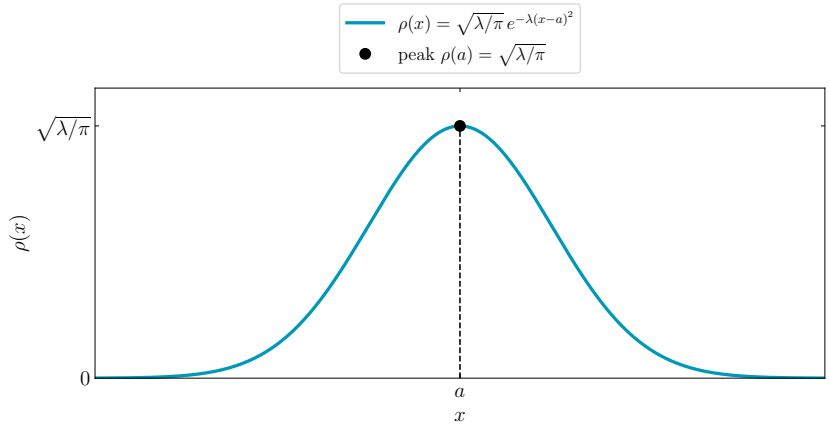
Next we have

$$\begin{aligned}\langle x^2 \rangle &= \int_{-\infty}^{\infty} Ax^2 e^{-\lambda(x-a)^2}\, dx \\ &= A\left[\int_{-\infty}^{\infty} u^2 e^{-\lambda u^2}\, du + 2a\int_{-\infty}^{\infty} u e^{-\lambda u^2}\, du + \int_{-\infty}^{\infty} a^2 e^{-\lambda u^2}\, du\right] \\ &= \frac{1}{2\lambda} + a^2\end{aligned}$$

where we can conclude

$$\sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{1}{\sqrt{2\lambda}}$$

(c)



1.4 - Normalization

Problem: 1.4

At time $t = 0$ a particle is represented by the wave function

$$\Psi(x,0) = \begin{cases} A\dfrac{x}{a}, & \text{if } 0 \le x \le a, \\ A\dfrac{(b-x)}{(b-a)}, & \text{if } a \le x \le b, \\ 0, & \text{otherwise} \end{cases}$$

where A, a, and b are constants.

(a) Normalize Ψ (that is, find A, in terms of a and b).

(b) Sketch $\Psi(x,0)$, as a function of x.

(c) Where is the particle most likely to be found, at $t = 0$?

(d) What is the probability of finding the particle to the left of a? Check your result in the limiting cases $b = a$ and $b = 2a$.

(e) What is the expectation value of x?

(a) Normalizing the square-integrable wave function requires

$$\begin{aligned} \int_0^b |\Psi(x,0)|^2\,dx &= \int_0^b \Psi^*\Psi\,dx \\ &= |A|^2\left[\int_0^a \left(\frac{x}{a}\right)^2 dx + \int_a^b \left(\frac{b-x}{b-a}\right)^2 dx\right] \\ &= 1 \end{aligned}$$

We start by computing the integrals, which gets you

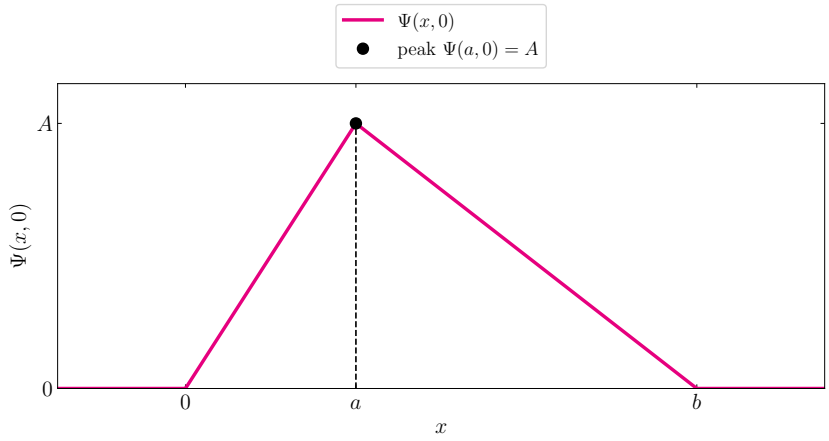
$$\begin{aligned} \int_0^a \left(\frac{x}{a}\right)^2 dx + \int_a^b \left(\frac{b-x}{b-a}\right)^2 dx &= \frac{x^3}{3a^2}\bigg|_0^a - \frac{(b-x)^3}{3(b-a)^2}\bigg|_a^b \\ &= \frac{a}{3} + \frac{b-a}{3} \\ &= \frac{b}{3} \end{aligned}$$

and leads to

$$|A|^2 = \frac{3}{b} \quad \Longrightarrow \quad |A| = A = \sqrt{\frac{3}{b}}$$

where we select the positive root by convention.

(b)



(c) From the plot in part (b), the particle is most likely to be found at $x = a$; since $\Psi(x,0) \geq 0$, it follows that $\arg\max_x \Psi(x,0) = \arg\max_x |\Psi(x,0)|^2 = a$.

(d) The probability that the particle is found within $0 \leq x \leq a$ is given by integrating $|\Psi(x,0)|^2$ over that interval:

$$P(0 \leq x \leq a) = \frac{3}{b}\int_0^a \left(\frac{x}{a}\right)^2 dx = \frac{3}{b}\left(\frac{x^3}{3a^2}\right)\Bigg|_0^a = \frac{a}{b}$$

When $a = b$, $P = 1$, which is physically sensible as the particle must fall within $[0, b]$. When $a = b/2$, $P = 1/2$, and $\Psi(x,0)$ is an isosceles triangle bisected down $x = b/2$.

(e) Derive

$$\begin{aligned}
\langle x \rangle &= \int_a^b x(\Psi^*\Psi)\, dx \\
&= \frac{3}{b}\left[\int_0^a x\left(\frac{x}{a}\right)^2 dx + \int_a^b x\left(\frac{b-x}{b-a}\right)^2 dx\right] \\
&= \frac{3}{b}\left[\frac{x^4}{4a^2}\Bigg|_0^a - \frac{1}{(b-a)^2}\left(\frac{b(b-x)^3}{3} - \frac{(b-x)^4}{4}\right)\Bigg|_a^b\right] \\
&= \frac{3}{b}\left(\frac{a^2}{4} + \frac{b(b-a)}{3} - \frac{(b-a)^2}{4}\right) \\
&= \frac{1}{4b}\left(3a^2 + 4b^2 - 4ab - 3b^2 + 6ab - 3a^2\right) \\
&= \frac{2ab + b^2}{4b} \\
&= \frac{2a + b}{4}
\end{aligned}$$

Problem: 1.5

Consider the wave function

$$\Psi(x,t) = Ae^{-\lambda|x|}e^{-i\omega t}$$

where A, λ, and ω are positive real constants. (We'll see in Chapter 2 what potential (V) actually produces such a wave function.)

(a) Normalize Ψ.

(b) Determine the expectation values of x and x^2.

(c) Find the standard deviation of x. Sketch the graph of $|\Psi|^2$, as a function of x, and mark the points $(\langle x\rangle + \sigma)$ and $(\langle x\rangle - \sigma)$, to illustrate the sense in which σ represents the "spread" in x. What is the probability that the particle would be found outside this range?

(a) We require

$$\begin{aligned}\int_{-\infty}^{\infty} |\Psi(x,t)|^2\,dx &= \int_{-\infty}^{\infty} \Psi^*\Psi\,dx \\ &= |A|^2 \int_{-\infty}^{\infty} \left(e^{-\lambda|x|}e^{-i\omega t}\right)\left(e^{-\lambda|x|}e^{i\omega t}\right)\,dx \\ &= 2|A|^2 \int_0^{\infty} e^{-2\lambda x}\,dx \\ &= 1\end{aligned}$$

Solving for A yields

$$\begin{aligned}2|A|^2 \int_0^{\infty} e^{-2\lambda x}\,dx = -|A|^2 \frac{e^{-2\lambda x}}{\lambda}\Big|_0^{\infty} \\ = \frac{|A|^2}{\lambda} = 1 \\ \implies A = \sqrt{\lambda}\end{aligned}$$

where by premise, A is positive.

(b) Compute

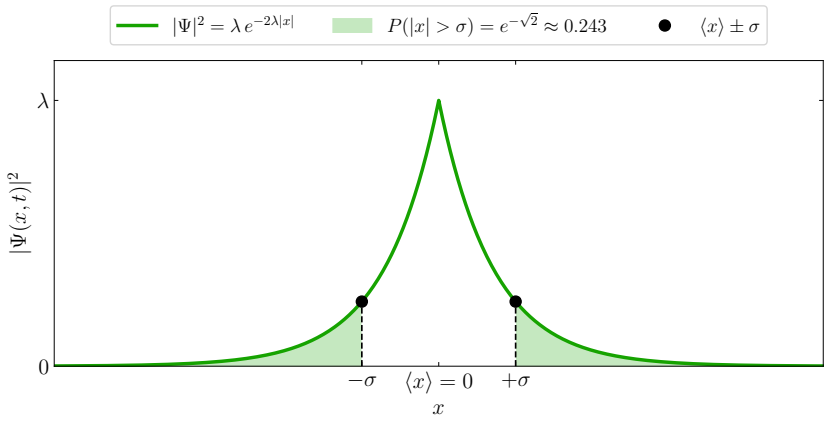
$$\begin{aligned}
\langle x \rangle &= \lambda \int_{-\infty}^{\infty} x \Big(e^{-\lambda|x|} e^{-i\omega t} \Big) \Big(e^{-\lambda|x|} e^{i\omega t} \Big) \, dx \\
&= \lambda \int_{-\infty}^{\infty} x e^{-2\lambda|x|} \, dx \\
&= 0 \\
\langle x^2 \rangle &= \lambda \int_{-\infty}^{\infty} x^2 \Big(e^{-\lambda|x|} e^{-i\omega t} \Big) \Big(e^{-\lambda|x|} e^{i\omega t} \Big) \, dx \\
&= 2\lambda \int_{0}^{\infty} x^2 e^{-2\lambda x} \, dx \\
&= -e^{-2\lambda x} \left(x^2 + \frac{x}{\lambda} + \frac{1}{2\lambda^2} \right) \Bigg|_0^{\infty} \\
&= \frac{1}{2\lambda^2}
\end{aligned}$$

where antisymmetry about the origin causes the mean $\langle x \rangle$ to vanish and integration by parts leads to the result for $\langle x^2 \rangle$.

(c) Derive the standard deviation as

$$\sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{1}{\sqrt{2}\lambda}$$

Plotting the probability density:



It is easy to see that appealing to the symmetry of the density, the probability that the particle is found beyond one standard deviation of the mean is

$$\begin{aligned}
P(\sigma \leq x) + P(x \leq -\sigma) &= 2\lambda \int_{+\sigma}^{\infty} e^{-2\lambda x} \, dx \\
&= -e^{-2\lambda x} \Big|_{+\sigma}^{\infty} \\
&= e^{-\sqrt{2}} \approx 0.243
\end{aligned}$$

1.5 - Momentum

Problem: 1.6

Why can't you do integration-by-parts directly on the middle expression in Equation 1.29 – pull the time derivative over onto x, note that $\partial x/\partial t = 0$, and conclude that $d\langle x\rangle/dt = 0$?

Equation 1.29 says

$$\frac{d\langle x\rangle}{dt} = \int x\frac{\partial}{\partial t}|\Psi|^2\,dx = \frac{i\hbar}{2m}\int x\frac{\partial}{\partial x}\left(\Psi^*\frac{\partial\Psi}{\partial x} - \frac{\partial\Psi^*}{\partial x}\Psi\right)dx$$

Consider the product rule

$$\frac{\partial}{\partial t}\left(x|\Psi|^2\right) = |\Psi|^2\frac{\partial x}{\partial t} + x\frac{\partial|\Psi|^2}{\partial t} = x\frac{\partial}{\partial t}|\Psi|^2$$

In this case, x is not parameterized by t, so $\partial x/\partial t = 0$, but the second term still remains. Our variable of integration, however, is x, *not* t. If we were to integrate over x:

$$\frac{d\langle x\rangle}{dt} = \int\frac{\partial}{\partial t}\left(x|\Psi|^2\right)dx = \int x\frac{\partial}{\partial t}|\Psi|^2\,dx$$

which merely returns us to the middle part of Equation 1.29. On this alone, we would not be able to conclude that the last term is zero, and as such, we cannot conclude $d\langle x\rangle/dt = 0$.

Problem: 1.7

Calculate $d\langle p\rangle/dt$. *Answer*:

$$\frac{d\langle p\rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle$$

Equations 1.32 (or the first part of 1.33) and 1.38 are instances of **Ehrenfest's theorem**, which tells us that *expectation values obey classical laws*.

Compute, by way of Leibniz's rule

$$\begin{aligned}
\frac{d\langle p\rangle}{dt} &= -i\hbar\frac{d}{dt}\int \Psi^*\frac{\partial\Psi}{\partial x}\,dx = -i\hbar\int \frac{\partial}{\partial t}\left(\Psi^*\frac{\partial\Psi}{\partial x}\right)\,dx \\
&= -i\hbar\int\left(\frac{\partial\Psi^*}{\partial t}\frac{\partial\Psi}{\partial x} + \Psi^*\underbrace{\frac{\partial}{\partial t}\frac{\partial\Psi}{\partial x}}_{=\frac{\partial}{\partial x}\frac{\partial\Psi}{\partial t}}\right)dx \\
&= -i\hbar\int\left[\left(-\frac{i\hbar}{2m}\frac{\partial^2\Psi^*}{\partial x^2} + \frac{i}{\hbar}V\Psi^*\right)\frac{\partial\Psi}{\partial x} + \Psi^*\frac{\partial}{\partial x}\left(\frac{i\hbar}{2m}\frac{\partial^2\Psi}{\partial x^2} - \frac{i}{\hbar}V\Psi\right)\right]dx \\
&= -\frac{\hbar^2}{2m}\int\left(\frac{\partial^2\Psi^*}{\partial x^2}\frac{\partial\Psi}{\partial x} - \Psi^*\frac{\partial^3\Psi}{\partial x^3}\right)dx + \int\left(V\Psi^*\frac{\partial\Psi}{\partial x} - \Psi^*\frac{\partial}{\partial x}(V\Psi)\right)dx \\
&= \int\left[\underbrace{\left(V\Psi^*\frac{\partial\Psi}{\partial x} - V\Psi^*\frac{\partial\Psi}{\partial x}\right)}_{=0} - \Psi^*\Psi\frac{\partial V}{\partial x}\right]dx \\
&= -\int\frac{\partial V}{\partial x}|\Psi|^2\,dx \\
&= \left\langle -\frac{\partial V}{\partial x}\right\rangle
\end{aligned}$$

where in going from the fourth line to the fifth we use the fact that $\Psi, \Psi^* \to 0$ along with their first derivative as $x \to \pm\infty$.

Problem: 1.8

Suppose you add a constant V_0 to the potential energy (by "constant" I mean independent of x as well as t). In *classical* mechanics this doesn't change anything, but what about *quantum mechanics*? Show that the wave function picks up a time-dependent phase factor: $\exp(-iV_0t/\hbar)$. What effect does this have on the expectation value of a dynamical variable?

In essence, we wish to prove that for the Schrödinger equation with an added constant potential energy term

$$i\hbar\frac{\partial\tilde{\Psi}}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\tilde{\Psi}}{\partial x^2} + (V + V_0)\,\tilde{\Psi}$$

we have a solution with a time-dependent phase factor

$$\tilde{\Psi} = \Psi e^{-iV_0 t/\hbar}$$

First compute

$$\frac{\partial \tilde{\Psi}}{\partial t} = \frac{\partial \Psi}{\partial t} e^{-iV_0 t/\hbar} - \frac{iV_0}{\hbar} \Psi e^{-iV_0 t/\hbar}$$
$$\frac{\partial^2 \tilde{\Psi}}{\partial x^2} = \frac{\partial^2 \Psi}{\partial x^2} e^{-iV_0 t/\hbar}$$

Then substitute into Schrödinger's equation and cancel the $e^{-iV_0 t/\hbar}$ terms:

$$i\hbar \frac{\partial \Psi}{\partial t} + V_0 \Psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + (V + V_0)\, \Psi$$

and canceling the $V_0 \Psi$ terms restores the original Schrödinger's equation, as desired. Referring to equation 1.36 for the expectation value of any dynamical quantity:

$$\langle Q(x,p) \rangle = \int \Psi^* Q\left(x, \frac{\hbar}{i} \frac{\partial}{\partial x}\right) \Psi \, dx$$

And substituting $\tilde{\Psi}$, bearing in mind that the time-dependent phase factor is exclusively in t and thus treated as a constant in the integral over x, we find

$$\begin{aligned} \langle Q(x,p) \rangle &= \int \tilde{\Psi}^* Q\left(x, \frac{\hbar}{i} \frac{\partial}{\partial x}\right) \tilde{\Psi} \, dx \\ &= \int \underbrace{\Psi^* e^{iV_0 t/\hbar}} Q\left(x, \frac{\hbar}{i} \frac{\partial}{\partial x}\right) \underbrace{\Psi e^{-iV_0 t/\hbar}} \, dx \\ &= \int \Psi^* Q\left(x, \frac{\hbar}{i} \frac{\partial}{\partial x}\right) \Psi \, dx \end{aligned}$$

meaning that expectation values are **unchanged** by the presence of a constant potential energy.

1.6 - The Uncertainty Principle

Problem: 1.9

A particle of mass m is in the state

$$\Psi(x,t) = Ae^{-a\left[\left(mx^2/\hbar\right)+it\right]}$$

where A and a are positive real constants.

(a) Find A.

(b) For what potential energy function $V(x)$ does Ψ satisfy the Schrödinger equation?

(c) Calculate the expectation values of x, x^2, p, and p^2.

(d) Find σ_x and σ_p. Is their product consistent with the uncertainty principle?

(a) Normalize to derive

$$\begin{aligned}\int_{-\infty}^{\infty} \Psi^*\Psi\, dx &= |A|^2 \int_{-\infty}^{\infty} \left(e^{-a\left[\left(mx^2/\hbar\right)-it\right]}\right)\left(e^{-a\left[\left(mx^2/\hbar\right)+it\right]}\right) dx \\ &= |A|^2 \int_{-\infty}^{\infty} e^{-2amx^2/\hbar}\, dx \\ &= |A|^2 \sqrt{\frac{\pi\hbar}{2am}} = 1 \\ \implies A &= \left(\frac{2am}{\pi\hbar}\right)^{1/4}\end{aligned}$$

(b) Compute

$$\begin{aligned}\frac{\partial\Psi}{\partial t} &= -ai\Psi \\ \frac{\partial\Psi}{\partial x} &= -\frac{2am}{\hbar}x\Psi \\ \frac{\partial^2\Psi}{\partial x^2} &= \left(-\frac{2am}{\hbar} + \frac{4a^2m^2}{\hbar^2}x^2\right)\Psi\end{aligned}$$

Substituting into Schrödinger's equation produces

$$a\hbar\Psi = \left(a\hbar - 2a^2mx^2\right)\Psi + V(x)\,\Psi$$

Canceling Ψ throughout and rearranging to solve $V(x)$, we have

$$V(x) = 2a^2mx^2$$

(c) Using the result of problem 1.8, we need not consider the phase factor.

From equation 1.36 the expectations are

$$\begin{aligned}
\langle x\rangle &= \int_{-\infty}^{\infty} \Psi^* x \Psi \, dx = |A|^2 \int_{-\infty}^{\infty} x e^{-\alpha x^2} \, dx \\
&= 0 \\
\langle x^2\rangle &= \int_{-\infty}^{\infty} \Psi^* x^2 \Psi \, dx = |A|^2 \int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} \, dx = 2|A|^2 \int_0^{\infty} x^2 e^{-\alpha x^2} \, dx \\
&= \frac{|A|^2}{2\alpha}\sqrt{\frac{\pi}{\alpha}} \\
\langle p\rangle &= \int_{-\infty}^{\infty} \Psi^* \frac{\hbar}{i}\frac{\partial \Psi}{\partial x} \, dx = -\frac{2|A|^2 am}{i} \int_{-\infty}^{\infty} x e^{-\alpha x^2} \, dx \\
&= 0 \\
\langle p^2\rangle &= -\hbar^2 \int_{-\infty}^{\infty} \Psi^* \frac{\partial^2 \Psi}{\partial x^2} \, dx \\
&= -|A|^2\hbar^2 \int_{-\infty}^{\infty} \left(-\alpha + \alpha^2 x^2\right) e^{-\alpha x^2} \, dx \\
&= |A|^2 \left[\alpha\hbar^2 \int_{-\infty}^{\infty} e^{-\alpha x^2} \, dx - 2\alpha^2\hbar^2 \int_0^{\infty} x^2 e^{-\alpha x^2} \, dx\right] \\
&= |A|^2 \left[\alpha\hbar^2\sqrt{\frac{\pi}{\alpha}} - \frac{\alpha\hbar^2}{2}\sqrt{\frac{\pi}{\alpha}}\right] \\
&= \frac{|A|^2\alpha\hbar^2}{2}\sqrt{\frac{\pi}{\alpha}}
\end{aligned}$$

where $\alpha = 2am/\hbar$. Simplifying the nonzero expecations gives us

$$\begin{aligned}
\langle x^2\rangle &= \frac{1}{2}\left(\frac{2am}{\pi\hbar}\right)^{1/2}\left(\frac{\hbar}{2am}\right)^{3/2}\sqrt{\pi} = \frac{\hbar}{4am} \\
\langle p^2\rangle &= \frac{1}{2}\left(\frac{2am}{\pi\hbar}\right)^{1/2}\left(\frac{2am}{\hbar}\right)^{1/2}\hbar^2\sqrt{\pi} = am\hbar
\end{aligned}$$

(d) Since $\langle x\rangle = \langle p\rangle = 0$,

$$\begin{aligned}
\sigma_x &= \sqrt{\frac{\hbar}{4am}} \\
\sigma_p &= \sqrt{am\hbar}
\end{aligned}$$

We can conclude

$$\sigma_x\sigma_p = \frac{\hbar}{2}$$

which is beautifully consistent with the uncertainty principle. This is the wave function corresponding to the quantum harmonic oscillator in its ground-state (more on this in chapter 2).

Further Problems for Chapter 1

Problem: 1.10

Consider the first 25 digits in the decimal expansion of π (3, 1, 4, 1, 5, 9, ...).

(a) If you selected one number at random, from this set, what are the probabilities of getting each of the 10 digits?

(b) What is the most probable digit? What is the median digit? What is the average value?

(c) Find the standard deviation for this distribution.

Digit j	$P(j)$
0	0
1	2/25
2	3/25
3	5/25
4	3/25
5	3/25
6	3/25
7	1/25
8	2/25
9	3/25

(a)

(b) The most probable digit is 3, with $P(3) = 1/5$, with median digit 4, and average value

$$\begin{aligned}\langle j \rangle &= \sum jP(j) \\ &= \frac{1\cdot 2 + 2\cdot 3 + 3\cdot 5 + 4\cdot 3 + 5\cdot 3 + 6\cdot 3 + 7\cdot 1 + 8\cdot 2 + 9\cdot 3}{25} \\ &= \frac{118}{25} = 4.72\end{aligned}$$

(c) Compute the mean of j^2:

$$\begin{aligned}\langle j^2 \rangle &= \sum j^2 P(j) \\ &= \frac{1^2\cdot 2 + 2^2\cdot 3 + 3^2\cdot 5 + 4^2\cdot 3 + 5^2\cdot 3 + 6^2\cdot 3 + 7^2\cdot 1 + 8^2\cdot 2 + 9^2\cdot 3}{25} \\ &= \frac{710}{25} = 28.4\end{aligned}$$

Then the standard deviation is

$$\sigma = \sqrt{\langle j^2 \rangle - \langle j \rangle^2} = \sqrt{28.4 - 4.72^2} = \sqrt{6.1216} \approx 2.474$$

Problem: 1.11

The needle on a broken car speedometer is free to swing, and bounces perfectly off the pins on either end, so that if you give it a flick it is equally likely to come to rest at any angle between 0 and π.

(a) What is the probability density, $\rho(\theta)$? *Hint*: $\rho(\theta)\, d\theta$ is the probability that the needle will come to rest between θ and $(\theta + d\theta)$. Graph $\rho(\theta)$ as a function of θ, from $-\pi/2$ to $3\pi/2$. (Of course, *part* of this interval is excluded, so ρ is zero there.) Make sure that the total probability is 1.

(b) Compute $\langle\theta\rangle$, $\langle\theta^2\rangle$, and σ, for this distribution.

(c) Compute $\langle\sin(\theta)\rangle$, $\langle\cos(\theta)\rangle$, and $\langle\cos^2(\theta)\rangle$.

(a) The density is given by

$$\rho(\theta) = \begin{cases} \frac{1}{\pi}, & 0 \leq \theta \leq \pi \\ 0, & -\pi/2 \leq \theta < 0, \pi < \theta \leq 3\pi/2 \end{cases}$$

(b)